DO NOW — RECALL

LMC branching and debugging

Name: _____ Class: _____ Date: _____

1. What does STA do?

2. What does LMC HLT do?

3. Why must a loop's counter change?

4. Full adder: inputs A=1, B=0, carry-in=1. Find sum and carry-out.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Copies ACC to mailbox.

2. Model answer

Stops execution.

3. Model answer

Without updating a tested state, the loop may not terminate.

4. Model answer

$1+0+1=10_2$; sum=0, carry-out=1.

Vocabulary — know these terms

BRA — Branch Always

BRZ — Branch if Zero

BRP — Branch if non-negative

mailbox — LMC memory cell

NEW LEARNING

Read, annotate and explain

Understand

BRP tests whether ACC is non-negative; BRZ tests zero; BRA is unconditional.

Trace

Load the counter, output it and subtract one. Test BRP immediately: save the new value and loop only when it is non-negative.

Explain

After outputting 0, SUB ONE underflows; BRP is not taken, so HLT runs without storing a negative mailbox value.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace BRZ when input is 0.

Predict first

INP sets ACC to 0

Check next state

BRZ ZERO takes branch

Check next state

LDA ONE loads 1

Check next state

OUT displays 1.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define BRA, BRZ and BRP and explain their effect on PC. [4]

Question 2

Input 3: trace INP; BRZ STOP; SUB ONE; OUT; HLT; ONE DAT 1; STOP HLT. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why an unconditional backward branch without a tested exit can run forever, and provide a safe counter-based solution. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An unconditional backward branch always terminates.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.